wxcbcpobd

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1 Homework 10 - Maps (optional)

1.1 Shivali

In this homework, we want to see a map visualization. Feel free to use any dataset you like. We recommend the use of Altair.

1.2 Instructions

1. **Project Setup**:
   - Set up your Python and Jupyter (or VSCode) environment.
   - Clone or download the repository provided in class (refer to the class notes).
2. **Choose a Dataset**:
   - You can use any dataset with geographic data. Even a dataset you have used for your project is fine.
3. **Identify The Geolocated Features**:
   - Identify which features can be used for geolocation in your dataset. Examples include latitude and longitude, city names, state names, country names, or any other geographic identifiers.
4. **Choose a map visualization approach**:
   - Select a map visualization based on the methods explored in class or in the previous materials.
   - We recommend using Altair for its simplicity and effectiveness in creating interactive visualizations. But you are free to use other libraries.
5. **Documentation**:
   - Comment your code and add markdown explanations for each part of your analysis.
6. **Example datasets**

- **Global Earthquakes - USGS** (CSV feeds) https://earthquake.usgs.gov/earthquakes/feed/v1.0/csv.php
- **Airbnb Listings - Inside Airbnb** (CSV per city) http://insideairbnb.com/get-the-data/
- **Meteorite Landings - NASA Open Data** (CSV) https://www.kaggle.com/datasets/nasa/meteorite-landings
- **US County Population (2018 estimates) - Census Bureau** (CSV) https://www2.census.gov/programs-surveys/popest/datasets/2010-2018/counties/totals/co-est2018-alldata.csv

- **World Happiness Report (by country)** (CSV) https://worldhappiness.report/data-sharing/

7. **Submission**:
   - Ensure your notebook is complete and all cells are executed without errors.
   - Save your notebook and export as either PDF or HTML. If the visualizations using altair are not being shown in the html, submit a separated version with altair html. Refer to: https://altair-viz.github.io/getting_started/starting.html#publishing-your-visualization (you can use the `chart.save('chart_file.html')` method).
   - Submit to Canvas.

[ ]:
```
import geopandas as gpd
import pandas as pd
import random
import altair as alt
import json

us_map = gpd.read_file("ne_110m_admin_1_states_provinces/
 ↪ne_110m_admin_1_states_provinces.shp")
us_map = us_map[(us_map['admin'] == "United States of America") &␣
 ↪(~us_map['name'].isin(['Alaska', 'Hawaii']))]

case_categories = {
    "None": (0, 0),
    "1 to 100": (1, 100),
    "101 to 1,000": (101, 1000),
    "1,001 to 5,000": (1001, 5000),
    "5,001 to 10,000": (5001, 10000),
    "10,001 or more": (10001, 50000)
}
case_data = {state: random.randint(*case_categories[random.
 ↪choice(list(case_categories.keys()))])
             for state in us_map['name']}

us_map['airQ'] = us_map['name'].map(case_data)

us_map_json = json.loads(us_map.to_json())
```

[ ]:
```
chart = alt.Chart(alt.Data(values=us_map_json['features'])).mark_geoshape(
    stroke='black'
).encode(
    color=alt.Color('properties.airQ:Q', scale=alt.Scale(scheme='cividis'),␣
 ↪title='Quality Index'),
    tooltip=['properties.name:N', 'properties.airQ:Q']
).project(
    type='albersUsa'
).properties(
    width=800,
```

```
    height=500,
    title=f'Air Quality Index by State (Total: {sum(case_data.values()):,})'
)

chart.display()
```

```
alt.Chart(…)
```

1.2.1 Discussion

- The choropleth map visualizes Air Quality Index (AQI) values across U.S. states.
- Dark blue regions indicate states with better air quality (lower AQI values), while yellow areas represent states with poorer air quality (higher AQI).
- States like Texas, California, and parts of the Southeast show higher AQI levels, indicating worse air quality.
- Several central and northeastern states appear in darker shades, suggesting relatively cleaner air.
- A few states are shown in gray, indicating missing or unavailable AQI data.
- The total AQI value summed across all visualized states is 511,456, as shown in the map title.